

AI CHATBOT FOR INTERNAL HELPDESK

CodeVedX Internship Project

Submitted By

Name: Aarti Prajapati

Enrollment Number: 21231258

Course: B.Tech CSE (AI & ML), 7th Semester

College: IIMT University, Meerut ,Uttar Pradesh

Email : aartiaiml3009@gmail.com

Project Theme

The theme of this project is to develop an AI-powered chatbot that automates responses to frequently asked employee queries. By using Natural Language Processing (NLP) and Machine Learning techniques, the chatbot understands user questions and provides relevant answers instantly through a Flask-based web application. The project demonstrates the practical application of AI in improving workplace support and reducing manual effort.

GitHub Repository

https://github.com/aartiprajapati-ai/CODEVEDX/tree/main/AI_Chobot_Helpdesk

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INTRODUCTION

The rapid growth of Artificial Intelligence (AI) and Natural Language Processing (NLP) has transformed the way organizations provide support and assistance to their employees. Many companies receive a large number of repetitive queries related to office policies, leave management, attendance, password resets, and other administrative services. Responding to these queries manually can be time-consuming and may reduce the efficiency of the support team.

The **AI Chatbot for Internal Helpdesk** is developed to automate the process of answering frequently asked employee questions. The chatbot uses **Natural Language Processing (NLP)** techniques to understand user queries and a **Machine Learning** model based on **TF-IDF Vectorization** and **Logistic Regression** to generate accurate responses. The application is deployed using the **Flask** framework, providing a simple and interactive web interface for users.

This project demonstrates the complete machine learning workflow, including dataset creation, text preprocessing, feature extraction, model training, model persistence using Joblib, and deployment as a web application. It not only improves response time but also reduces the workload on the support team by providing instant and consistent answers to common employee queries.

The primary objective of this project is to demonstrate the practical implementation of Artificial Intelligence, Machine Learning, and NLP in building an efficient and user-friendly internal helpdesk chatbot that can enhance organizational productivity and employee support.

PROBLEM STATEMENT

In many organizations, employees frequently ask similar questions regarding office timings, leave policies, attendance, password resets, salary slips, and other HR or IT-related services. Handling these repetitive queries manually requires significant time and effort from the support team, leading to increased workload, slower response times, and reduced operational efficiency.

As the number of employees grows, providing quick and consistent responses to every query becomes more challenging. Delays in resolving common issues can negatively affect employee productivity and overall workplace experience.

To address this challenge, there is a need for an intelligent and automated system that can instantly answer frequently asked questions without requiring continuous human intervention. An AI-powered chatbot can provide accurate, fast, and reliable responses, improving employee satisfaction while reducing the workload of HR and IT support teams.

The **AI Chatbot for Internal Helpdesk** is developed to solve this problem by using **Natural Language Processing (NLP)** and **Machine Learning** techniques to understand employee queries and deliver relevant responses through a simple Flask-based web application

PROJECT OBJECTIVE

The main objectives of the **AI Chatbot for Internal Helpdesk** project are:

- To develop an AI-powered chatbot for answering frequently asked employee queries.
- To automate responses for common HR and IT-related questions.
- To reduce the workload of the support team by minimizing repetitive manual responses.
- To implement Natural Language Processing (NLP) techniques for understanding user queries.
- To train a Machine Learning model using TF-IDF Vectorization and Logistic Regression.
- To build a simple and interactive web application using the Flask framework.
- To provide quick, accurate, and consistent responses to users.
- To demonstrate the practical application of Artificial Intelligence and Machine Learning in a real-world helpdesk system

EXPECTED OUTCOMES

- Employees can receive instant answers to frequently asked questions.
- Reduced response time for common support queries.
- Improved efficiency of the internal helpdesk system.
- Better understanding of NLP, Machine Learning, and Flask web development.
- A functional AI chatbot that can be extended with additional features in the future.

SCOPE OF THE PROJECT

The scope of the **AI Chatbot for Internal Helpdesk** is to provide an intelligent and automated solution for answering frequently asked employee queries within an organization. The chatbot is designed to understand user questions and deliver relevant responses using **Natural Language Processing (NLP)** and **Machine Learning** techniques.

The current version of the chatbot focuses on answering common questions related to office timings, leave applications, password resets, HR policies, and other internal helpdesk services. The application is developed using the **Flask** framework, making it lightweight, user-friendly, and easy to deploy.

The project can be further enhanced by integrating advanced AI technologies, voice-based interaction, multilingual support, database connectivity, user authentication, and cloud deployment. These improvements would make the chatbot suitable for large-scale organizations and enterprise-level applications.

Overall, this project demonstrates the practical implementation of Artificial Intelligence and Machine Learning to improve employee support services and reduce manual effort in handling repetitive queries.

LIMITATIONS

Although the chatbot performs well for predefined questions, it has certain limitations:

- It can answer only the questions available in the training dataset.
- It does not understand complex or ambiguous queries.
- The chatbot is currently designed for English-language inputs only.
- It does not connect to a live database or HR management system.
- Voice interaction and real-time learning features are not included.
- The chatbot cannot perform user authentication or access confidential employee information.

SOFTWARE REQUIREMENTS

The following software tools and technologies were used to develop the **AI Chatbot for Internal Helpdesk** project.

Software	Purpose
Python 3.x	Programming Language
Visual Studio Code	Code Editor
Flask	Web Application Framework
Pandas	Data Processing
Scikit-learn	Machine Learning Model
Joblib	Model Saving and Loading
HTML	Web Page Structure
CSS	User Interface Design
Google Chrome	Running the Web Application
Git & GitHub	Version Control and Project Hosting

HARDWARE REQUIREMENTS

The minimum hardware requirements for developing and running this project are:

Hardware	Specification
Processor	Intel Core i3 or above
RAM	4 GB or higher (8 GB Recommended)
Storage	500 MB Free Disk Space
Operating System	Windows 10 / Windows 11
Internet Connection	Required for GitHub and Package Installation

PYTHON LIBRARIES USED

The project uses the following Python libraries:

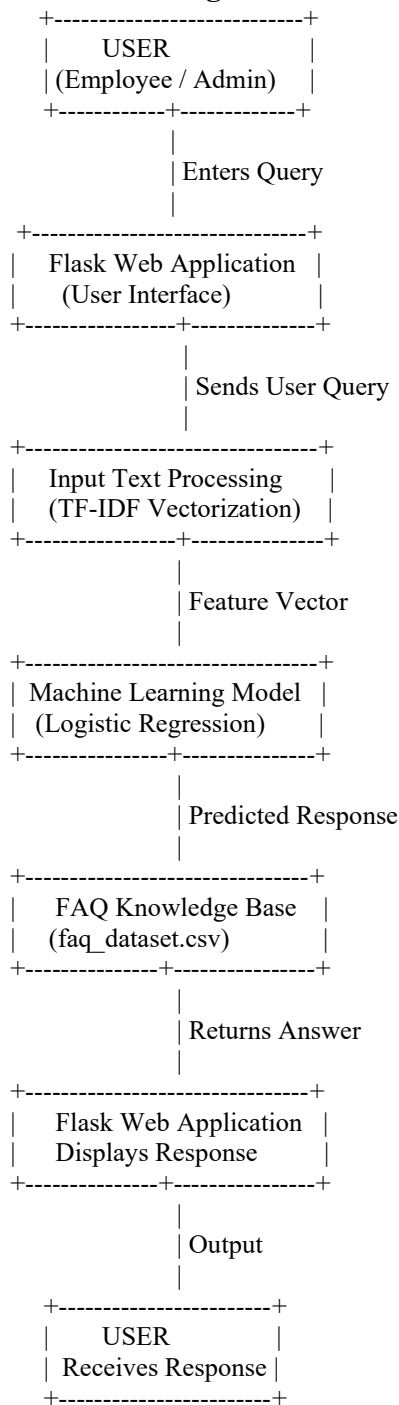
- Flask
- Pandas
- Scikit-learn
- Joblib

These libraries are installed using: `pip install -r requirements.txt`

SYSTEM ARCHITECTURE

The AI Chatbot for Internal Helpdesk follows a simple and efficient architecture. The user enters a query through the Flask web interface. The query is then preprocessed and converted into numerical features using the **TF-IDF Vectorizer**. These features are passed to the trained **Logistic Regression** model, which predicts the most appropriate response from the FAQ dataset. Finally, the chatbot displays the predicted response on the web page. The architecture ensures fast response time, easy deployment, and efficient handling of frequently asked employee queries.

System Architecture Diagram



WORKING OF THE SYSTEM

The chatbot works through the following steps:

1. The user enters a question in the chatbot interface.
2. Flask receives the input from the user.
3. The input text is converted into numerical features using the TF-IDF Vectorizer.
4. The trained Logistic Regression model processes these features.
5. The model predicts the most relevant response from the FAQ dataset.
6. Flask displays the predicted answer on the web page.
7. The user receives an instant response.

DATASET DESCRIPTION

The **AI Chatbot for Internal Helpdesk** uses a custom Frequently Asked Questions (FAQ) dataset containing common employee queries and their corresponding responses. The dataset is stored in CSV format and is used to train the Machine Learning model.

Each record in the dataset consists of a user question and its appropriate answer. During training, the chatbot learns the relationship between these questions and responses, enabling it to provide accurate answers when similar queries are asked by users.

The dataset can be easily updated by adding new question-answer pairs, allowing the chatbot to support additional topics without changing the application logic.

Dataset Information

Attribute	Description
Dataset Name	FAQ Dataset
File Format	CSV
File Name	faq_dataset.csv
Number of Columns	2
Input Feature	Question
Target Output	Answer

Dataset Schema

Column Name	Data Type	Description
Question	Text	Employee query entered by the user
Answer	Text	Corresponding response returned by the chatbot

Sample Dataset

Question	Answer
What are office timings?	Office timings are 9:30 AM to 6:30 PM.
How do I reset my password?	Contact the IT team or use the password reset portal.
How can I apply for leave?	Submit your leave request through the HR portal.
Who should I contact for payroll issues?	Please contact the HR or Payroll department.
How can I access the VPN?	Contact the IT department to enable VPN access.

DATA PREPROCESSING

Before training the model, the following preprocessing steps were performed:

- The dataset was loaded using **Pandas**.
- The **Question** column was selected as the input feature.
- The **Answer** column was used as the target output.
- The text data was converted into numerical vectors using the **TF-IDF Vectorizer**.
- The processed data was used to train the Logistic Regression model.

METHODOLOGY

The development of the **AI Chatbot for Internal Helpdesk** followed a structured Machine Learning workflow. The complete process is divided into several stages, from dataset preparation to chatbot deployment.

Step 1: Dataset Preparation

A custom FAQ dataset was created containing employee-related questions and their corresponding answers. The dataset was stored in CSV format and served as the foundation for training the chatbot.

Step 2: Data Loading

The dataset was loaded into Python using the **Pandas** library for further processing.

```
df = pd.read_csv("dataset/faq_dataset.csv")
```

Step 3: Text Preprocessing

The **Question** column was selected as the input feature, while the **Answer** column was used as the target output. The textual data was converted into numerical vectors using the **TF-IDF Vectorizer**, allowing the Machine Learning model to process the input.

Step 4: Model Training

The processed data was used to train a **Logistic Regression** model. This model learned the relationship between user questions and their corresponding responses.

```
model = LogisticRegression()  
model.fit(X_vectorized, y)
```

Step 5: Model Saving

After successful training, the Machine Learning model and the TF-IDF Vectorizer were saved using the **Joblib** library. Saving the model allows it to be reused without retraining every time the application starts.

```
joblib.dump(model, "model/chatbot_model.pkl")  
joblib.dump(vectorizer, "model/vectorizer.pkl")
```

Step 6: Flask Web Application

A Flask web application was developed to provide a simple and interactive user interface. Users can enter their questions through a web page, and the chatbot returns the predicted response.

Step 7: Response Prediction

When a user submits a question:

- The input text is converted into TF-IDF features.
- The trained Logistic Regression model predicts the most relevant response.
- Flask displays the response on the webpage.

PROJECT IMPLEMENTATION

The project consists of the following major components:

1. Dataset (faq_dataset.csv)

Stores all frequently asked questions and their corresponding answers used for training the chatbot.

2. Training Module (train_model.py)

Responsible for:

- Loading the dataset
- TF-IDF Vectorization
- Training the Logistic Regression model
- Saving the trained model and vectorizer

3. Web Application (app.py)

Responsible for:

- Loading the trained model
- Accepting user input
- Predicting the appropriate response
- Displaying the result using Flask

4. User Interface (templates/index.html)

Provides a clean and responsive interface where users can:

- Enter a question
- Click the **Ask Chatbot** button
- View the chatbot's response instantly

5. Model Files

The model folder contains:

- **chatbot_model.pkl** – Trained Machine Learning model
- **vectorizer.pkl** – Saved TF-IDF Vectorizer

These files are loaded by the Flask application to generate predictions.

RESULTS AND OUTPUT

The AI Chatbot for Internal Helpdesk was successfully developed and tested using a custom FAQ dataset. The chatbot is capable of understanding employee queries and providing relevant responses through a simple Flask-based web interface.

The Machine Learning model correctly predicts responses for common employee questions such as office timings, password reset, leave application, and payroll-related queries. The application provides fast, consistent, and user-friendly interaction, making it suitable for handling frequently asked questions within an organization.

The successful implementation of this project demonstrates the practical use of **Natural Language Processing (NLP)** and **Machine Learning** in automating internal support services.

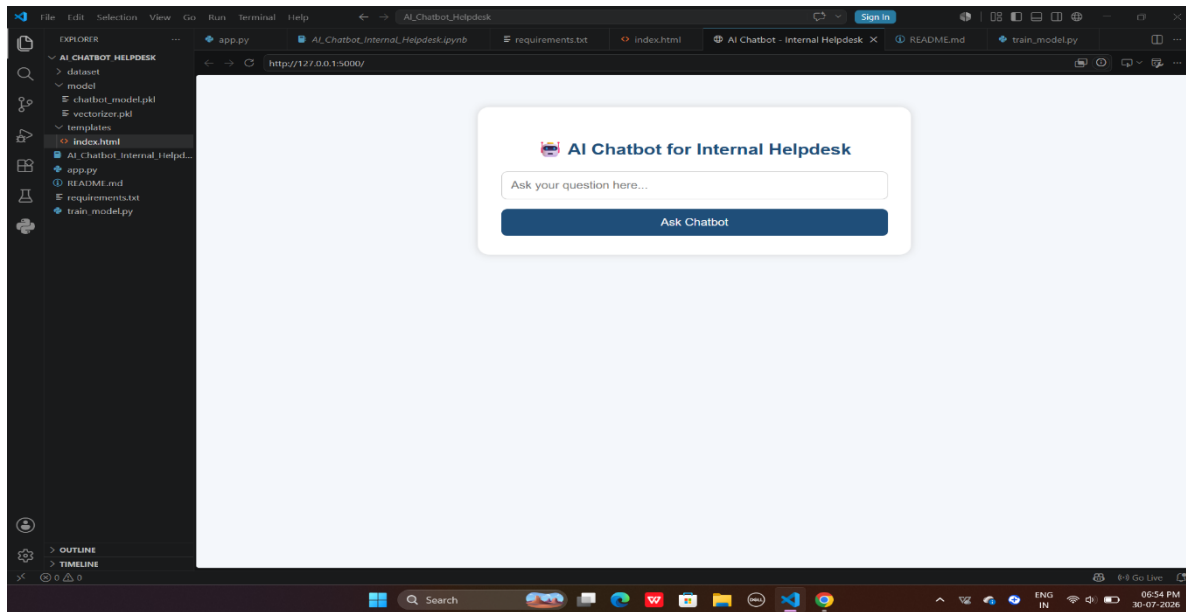
Output Screenshots

1. Home Page

Description:

The home page provides a simple interface where users can type their questions and submit them using the **Ask Chatbot** button.

Screenshot:

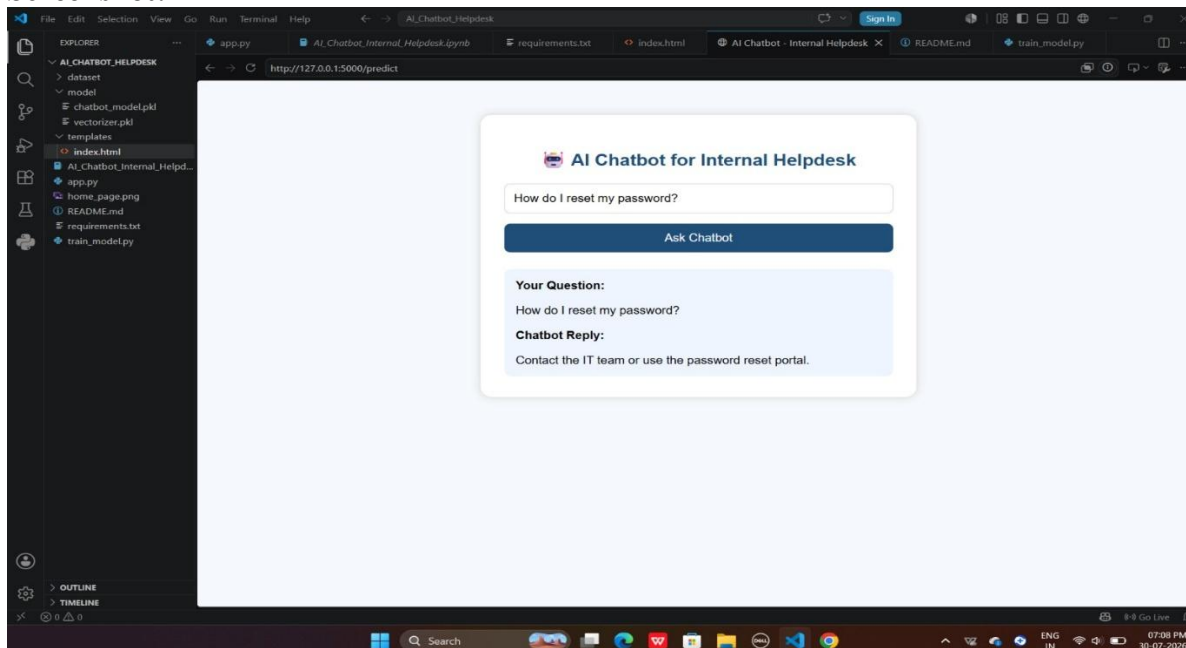


2. Password Reset Response

Description:

The chatbot successfully responds to password reset-related queries by providing the appropriate guidance.

Screenshot:

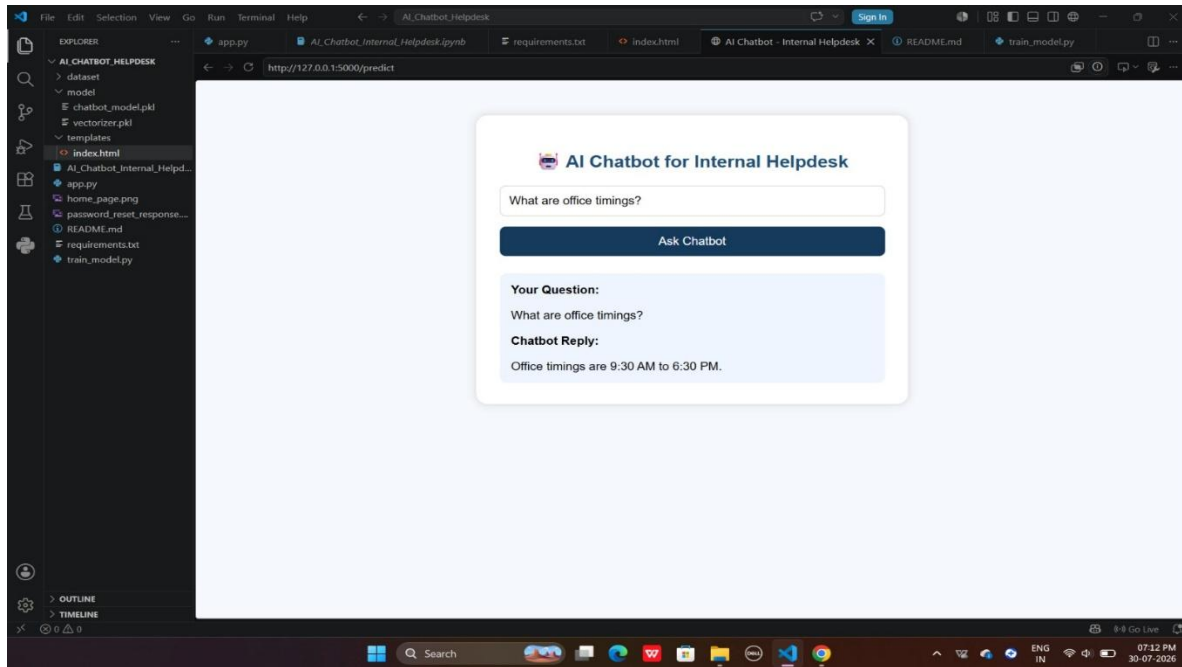


3. Office Timings Response

Description:

The chatbot correctly answers questions related to office working hours.

Screenshot:

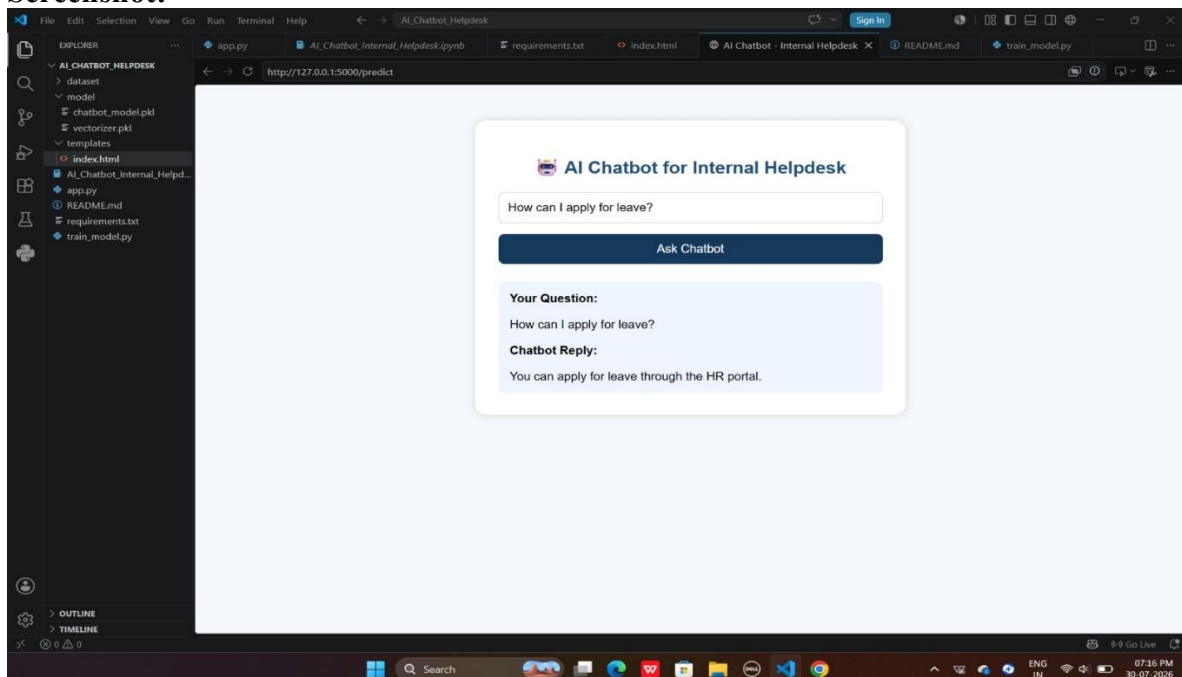


4. Leave Request Response

Description:

The chatbot provides guidance on the leave application process based on the trained FAQ dataset.

Screenshot:

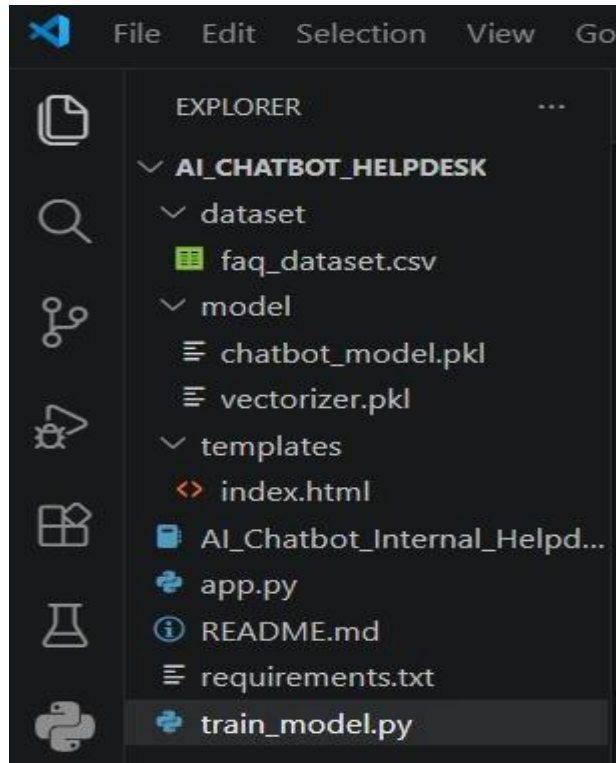


5. Project Structure

Description:

The project follows a well-organized folder structure, making it easy to understand, maintain, and extend.

Screenshot:

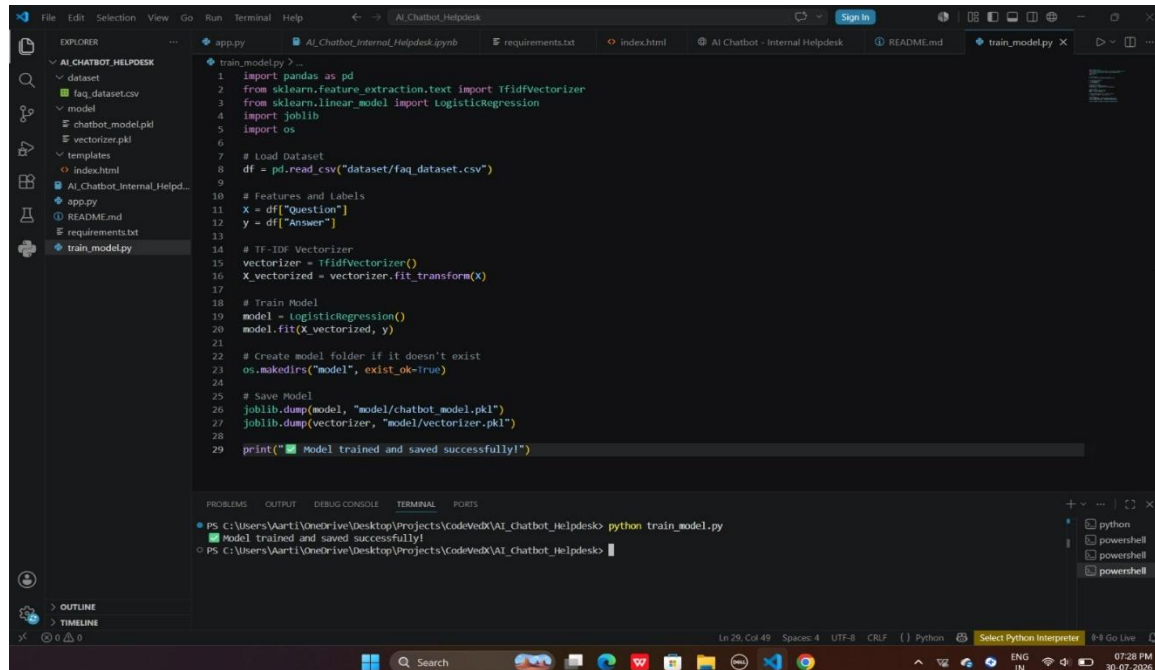


6. Model Training

Description:

The Machine Learning model was trained using the TF-IDF Vectorizer and Logistic Regression algorithm, and the trained model was saved using Joblib.

Screenshot:

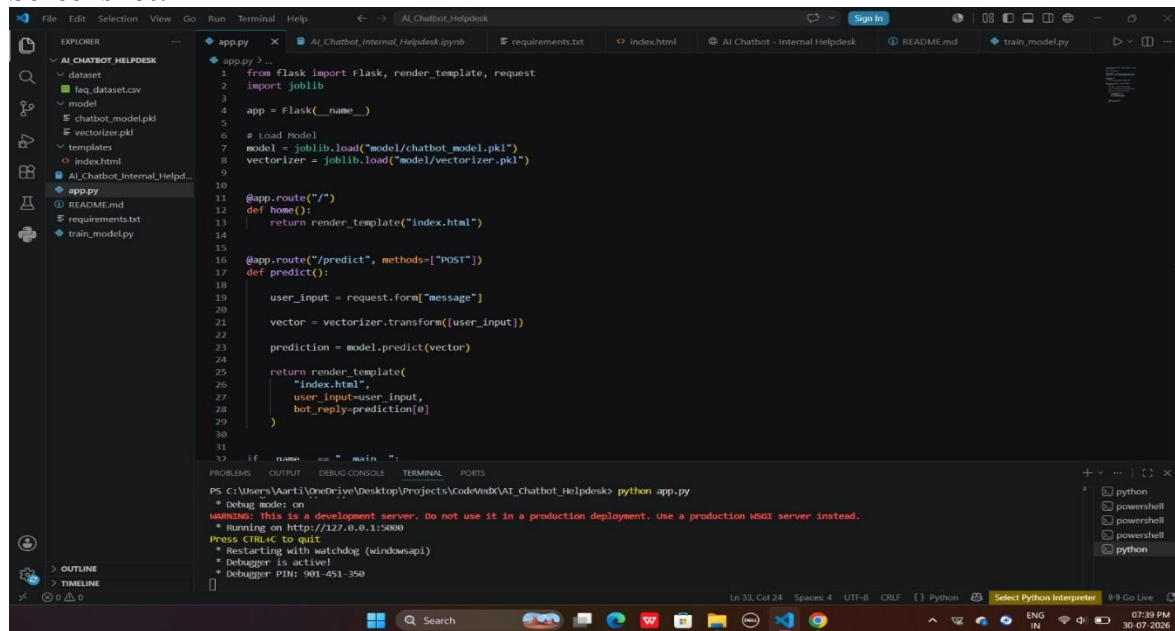


7. Flask Application Running

Description:

The Flask development server successfully hosts the chatbot application on the local system.

Screenshot:



The screenshot displays a Visual Studio Code editor window with the following components:

- Explorer Panel:** Shows the project structure for 'AI Chatbot Internal Helpdesk', including files like 'dataset', 'faq_dataset.csv', 'model', 'chatbot_model.pkl', 'vectorizer.pkl', 'templates', 'index.html', 'AI Chatbot Internal Helpdesk', 'app.py', 'README.md', 'requirements.txt', and 'train_model.py'.
- Editor Panel:** Displays the code for 'app.py'. The code imports Flask, render_template, request, and joblib. It defines a Flask app, loads a model and vectorizer, and sets up routes for a home page and a prediction endpoint. The prediction endpoint uses the loaded model and vectorizer to process user input and return a prediction.
- Terminal Panel:** Shows the command 'python app.py' being executed. The output indicates that the debug mode is on, the application is running on http://127.0.0.1:5000, and provides instructions for quitting (Press CTRL+C) and restarting with watchdog.

ADVANTAGES

- Provides instant responses to employee queries.
- Reduces the workload of HR and IT support teams.
- Improves response time and user experience.
- Easy to maintain and update with new FAQs.
- Lightweight and easy to deploy using Flask.
- Demonstrates practical implementation of AI and NLP concepts.

CONCLUSION

The **AI Chatbot for Internal Helpdesk** was successfully developed using **Natural Language Processing (NLP)**, **Machine Learning**, and the **Flask** web framework. The chatbot is capable of understanding employee queries and providing accurate responses based on a trained FAQ dataset.

The project demonstrates the complete machine learning lifecycle, including dataset preparation, text preprocessing, feature extraction using TF-IDF, model training with Logistic Regression, model persistence using Joblib, and deployment through a web-based interface.

The chatbot effectively reduces manual effort by automating responses to frequently asked questions, thereby improving response time and enhancing the overall user experience. This project also provided practical experience in applying Artificial Intelligence and Machine Learning techniques to solve a real-world business problem.

Overall, the project successfully achieved its objectives and serves as a strong foundation for developing more advanced AI-powered conversational systems.

FUTURE SCOPE

Although the current chatbot performs well for predefined employee queries, several enhancements can be implemented to improve its functionality and usability in real-world environments.

Some possible future improvements include:

- Integrating Large Language Models (LLMs) for more intelligent conversations.
- Adding voice-based interaction using Speech-to-Text and Text-to-Speech technologies.
- Supporting multiple languages for a wider range of users.
- Connecting the chatbot to a database for dynamic and real-time information retrieval.
- Implementing user authentication and role-based access control.
- Developing an admin dashboard to manage FAQ datasets easily.
- Deploying the chatbot on cloud platforms such as AWS, Azure, or Google Cloud.
- Integrating the chatbot with Microsoft Teams, Slack, or company intranet portals.
- Improving intent recognition using advanced Deep Learning and Transformer-based NLP models.

These enhancements would make the chatbot more scalable, intelligent, and suitable for enterprise-level deployment.

REFERENCES

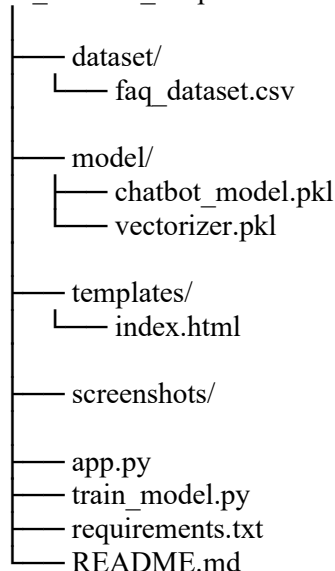
The following resources were referred to during the development of this project:

1. Python Official Documentation – <https://docs.python.org/>
2. Flask Documentation – <https://flask.palletsprojects.com/>
3. Scikit-learn Documentation – <https://scikit-learn.org/>
4. Pandas Documentation – <https://pandas.pydata.org/>
5. Joblib Documentation – <https://joblib.readthedocs.io/>
6. GitHub – <https://github.com/aartiprajapati-ai/>

APPENDIX

Appendix A – Project Folder Structure

AI_Chatbot_Helpdesk



Appendix B – Project Modules

- Dataset Preparation
- Text Preprocessing
- TF-IDF Vectorization
- Logistic Regression Model
- Model Saving using Joblib
- Flask Web Application
- User Interface
- Response Prediction

